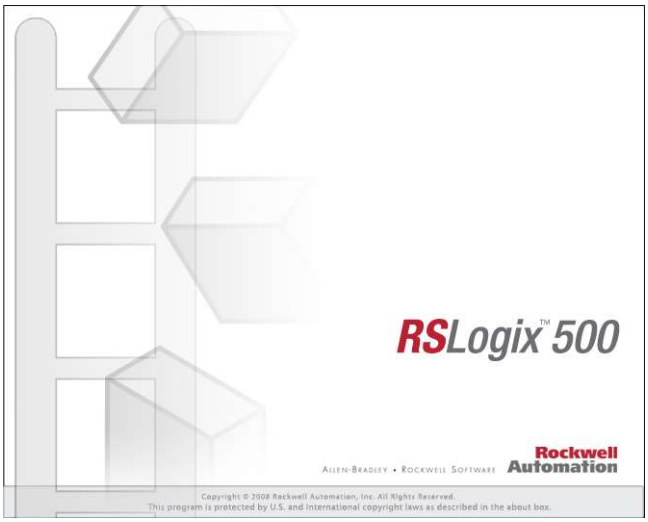
RSLogix 500 Project Report



Processor Information

Processor Type: Bul.1762 MicroLogix 1200 Series C (1 or 2 Comm Ports)

Processor Name: UNTITLED

Total Memory Used: 287 Instruction Words Used - 52 Data Table Words Used

Total Memory Left: 5409 Instruction Words Left

Program Files: 8

Data Files: 9

Program ID: fa6f

I/O Configuration

0	Bul.1762	MicroLogix 1200 Series C (1 or 2 C
1		
2		
3		
4		
5		
6		

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

Prog/HMI Port - Driver: DF1 Full Duplex
   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

Program File List

Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
MAIN	2	LADDER	9	No	218
SUB0	3	LADDER	9	No	278
SUB1	4	LADDER	6	No	137
SUB2	5	LADDER	5	No	79
SUB3	6	LADDER	9	No	278
SUB4	7	LADDER	8	No	193

Data File List

Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	12	4	O:3
INPUT	1	I	Global	No	12	4	I:3
STATUS	2	S	Global	No	0	66	S:65
BINARY	3	B	Global	No	1	1	B3:0
TIMER	4	T	Global	No	18	6	T4:5
COUNTER	5	C	Global	No	3	1	C5:0
CONTROL	6	R	Global	No	3	1	R6:0
INTEGER	7	N	Global	No	1	1	N7:0
FLOAT	8	F	Global	No	2	1	F8:0

subroutine

subroutine; selection 5 turn all off and resets/initialises; this program has sub0 that turns on lights from right to left and back; sub1 from the middle the lights turn on towards the righ and left ends and they merge again back to the centre; sub2 all led flashing.
OBSERVATION: all timers and counters must have unique address across all the files

Select subroutine
SELECT

reset to one
(default)

0000

I:0
0
Bul.1762

CTU
Count Up
Counter C5:0
Preset 8<
Accum 1<
CU
DN

reset mode selection to one (default behaviour): initialise to zero

reset to one
(default)

0001

EQU
Equal
Source A C5:0.ACC 1<
Source B 1 1<

MOV
Move
Source 0 0<
Dest O:0.0 0<

T4:0 RES
T4:1 RES
T4:5 RES

reset to one
(default)

0002

EQU
Equal
Source A C5:0.ACC 1<
Source B 2 2<

JSR
Jump To Subroutine
SBR File Number U:3

reset to one
(default)

0003

EQU
Equal
Source A C5:0.ACC 1<
Source B 3 3<

JSR
Jump To Subroutine
SBR File Number U:4

reset to one
(default)

0004

EQU
Equal
Source A C5:0.ACC 1<
Source B 4 4<

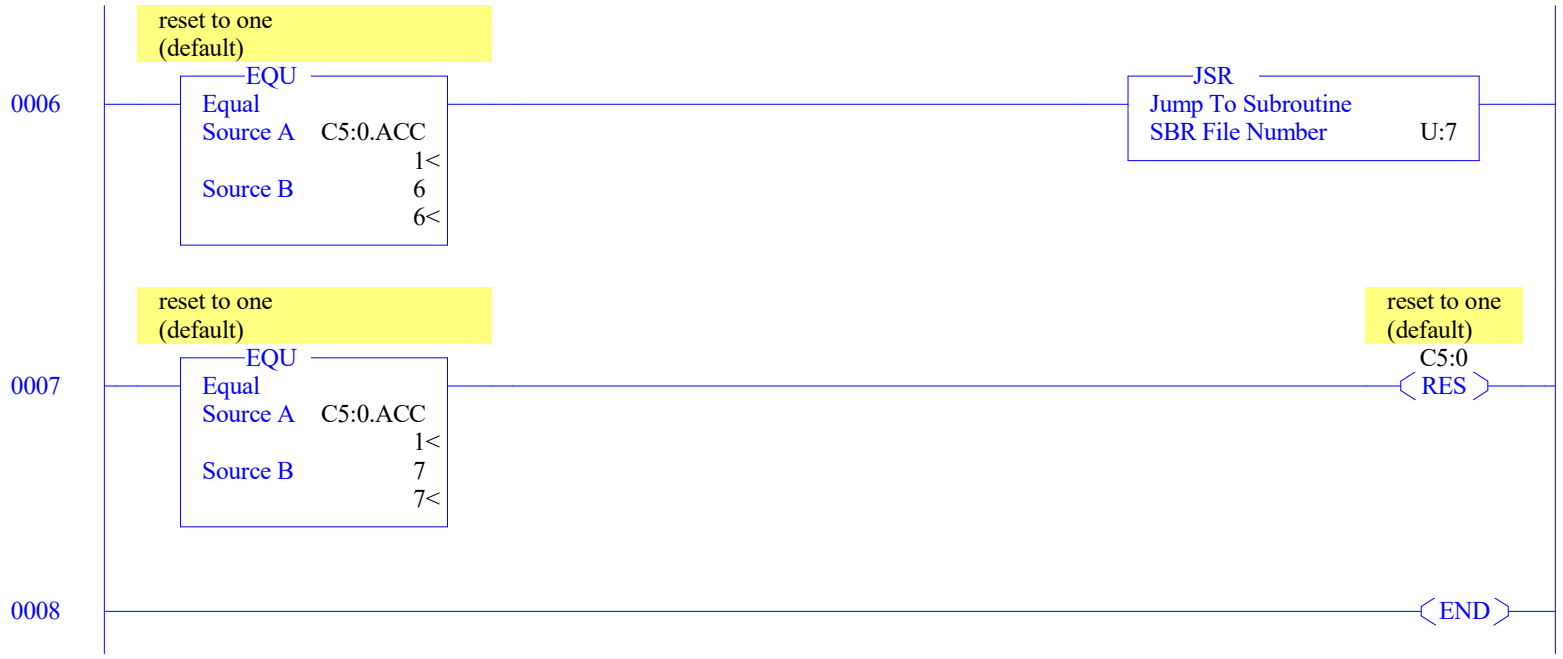
JSR
Jump To Subroutine
SBR File Number U:5

reset to one
(default)

0005

EQU
Equal
Source A C5:0.ACC 1<
Source B 5 5<

JSR
Jump To Subroutine
SBR File Number U:6


reset to one
(default)
0006
EQU
Equal
Source A C5:0.ACC
1<
Source B 6
6<
JSR
Jump To Subroutine
SBR File Number U:7
reset to one
(default)
reset to one
(default)
C5:0
0007
EQU
Equal
Source A C5:0.ACC
1<
Source B 7
7<
RES
0008
END

turn on right to left and back (solid one on per time)

0000

TON
Timer On Delay
Timer T4:0
Time Base 1.0
Preset 10<
Accum 0<
EN
DN

0001

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 1
1<

MOV
Move
Source 1
1<
Dest O:0.0
0<

0002

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 2
2<

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 10
10<

MOV
Move
Source 2
2<
Dest O:0.0
0<

0003

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 3
3<

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 9
9<

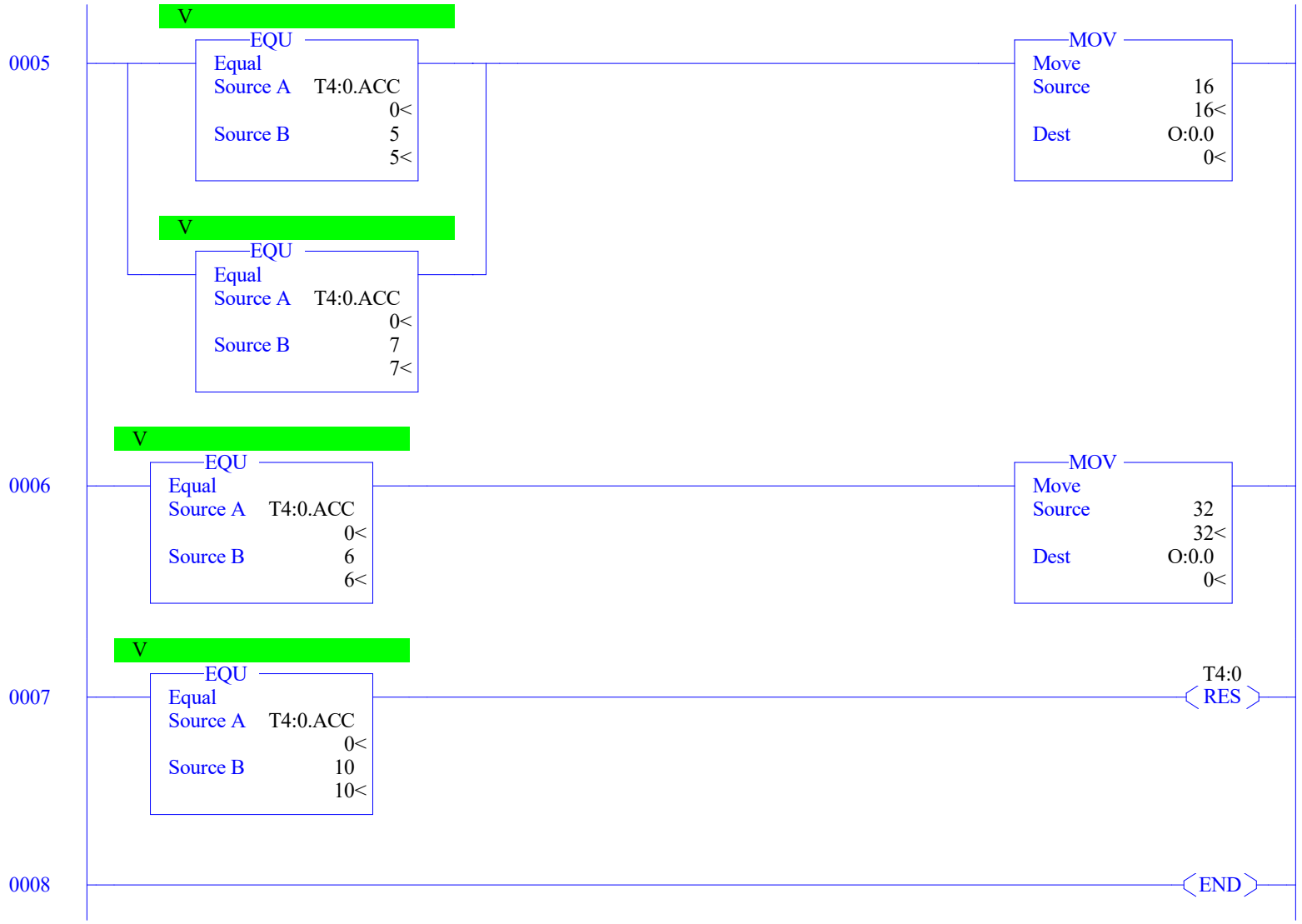
MOV
Move
Source 4
4<
Dest O:0.0
0<

0004

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 4
4<

V
EQU
Equal
Source A T4:0.ACC
0<
Source B 8
8<

MOV
Move
Source 8
8<
Dest O:0.0
0<


0005
V
EQU
Equal
Source A T4:0.ACC
0<
Source B 5
5<
V
EQU
Equal
Source A T4:0.ACC
0<
Source B 7
7<
MOV
Move
Source 16
16<
Dest O:0.0
0<
0006
V
EQU
Equal
Source A T4:0.ACC
0<
Source B 6
6<
MOV
Move
Source 32
32<
Dest O:0.0
0<
0007
V
EQU
Equal
Source A T4:0.ACC
0<
Source B 10
10<
T4:0
RES
0008
END

from center towards the right and left ends and back (two lights on per time)

0000

TON
Timer On Delay
Timer T4:1
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

turn on two middle lights (outp 3 and 4)

0001

EQU
Equal
Source A T4:1.ACC
0<
Source B 1
1<

MOV
Move
Source 12
12<
Dest O:0.0
0<

turn on outp 5 and 2

0002

EQU
Equal
Source A T4:1.ACC
0<
Source B 2
2<

MOV
Move
Source 18
18<
Dest O:0.0
0<

turn on two end lights (out 6 and 1)

0003

EQU
Equal
Source A T4:1.ACC
0<
Source B 3
3<

MOV
Move
Source 33
33<
Dest O:0.0
0<

turn on two middle lights (outp 3 and 4)

0004

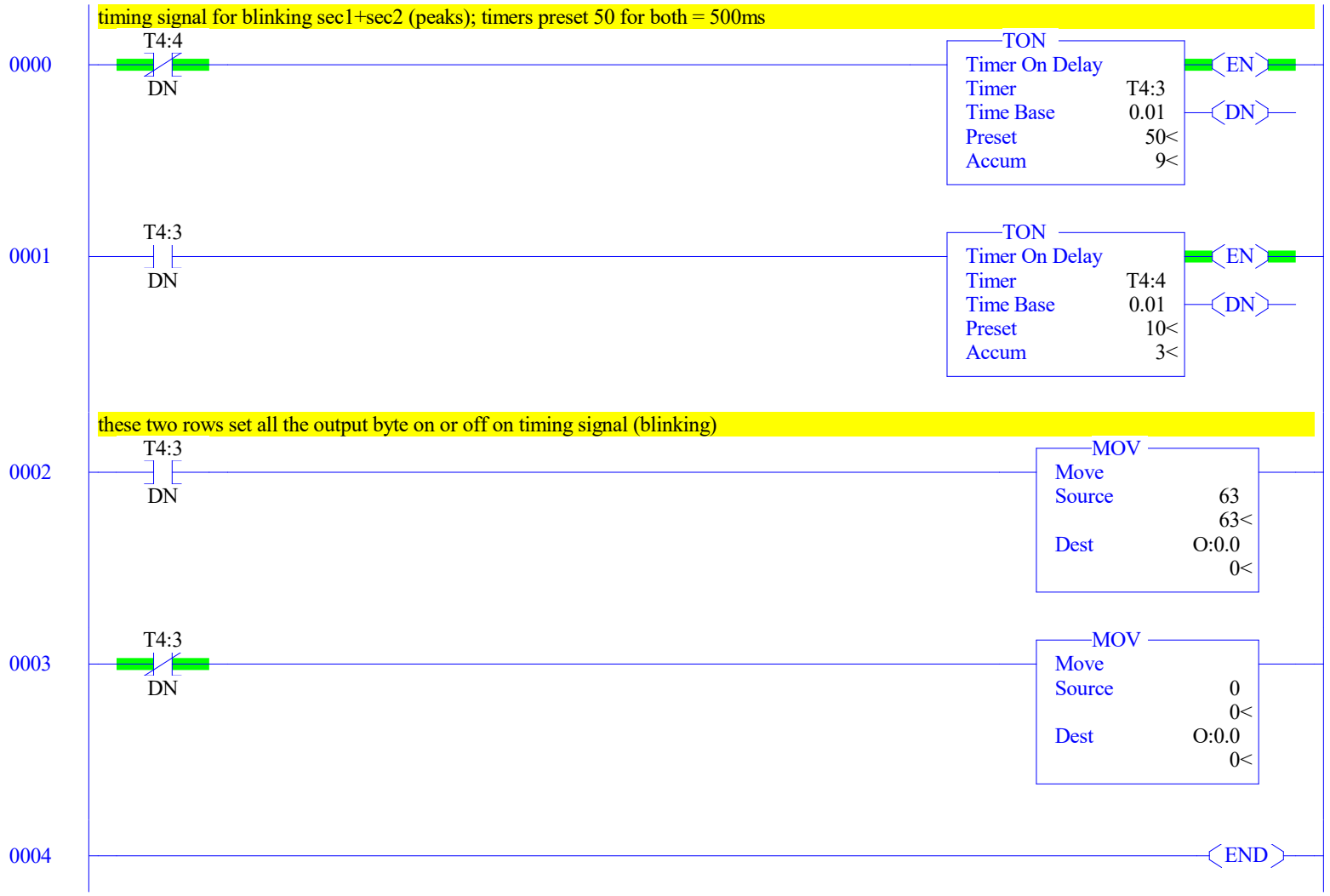
EQU
Equal
Source A T4:1.ACC
0<
Source B 4
4<

MOV
Move
Source 18
18<
Dest O:0.0
0<

T4:1
RES

0005

END


timing signal for blinking sec1+sec2 (peaks); timers preset 50 for both = 500ms
0000
T4:4
DN
TON
Timer On Delay
Timer T4:3
Time Base 0.01
Preset 50<
Accum 9<
EN
DN
0001
T4:3
DN
TON
Timer On Delay
Timer T4:4
Time Base 0.01
Preset 10<
Accum 3<
EN
DN
these two rows set all the output byte on or off on timing signal (blinking)
0002
T4:3
DN
MOV
Move
Source 63
63<
Dest O:0.0
0<
0003
T4:3
DN
MOV
Move
Source 0
0<
Dest O:0.0
0<
0004
END

turn on lights from right to left in a "latch style", similar to sub0

0000

TON
Timer On Delay
Timer T4:4
Time Base 1.0
Preset 10<
Accum 3<
EN
DN

turns on light1 only

0001

EQU
Equal
Source A T4:4.ACC 3<
Source B 1 1<

MOV
Move
Source 1 1<
Dest O:0.0 0<

0002

EQU
Equal
Source A T4:4.ACC 3<
Source B 2 2<

EQU
Equal
Source A T4:4.ACC 3<
Source B 10 10<

MOV
Move
Source 3 3<
Dest O:0.0 0<

0003

EQU
Equal
Source A T4:4.ACC 3<
Source B 3 3<

EQU
Equal
Source A T4:4.ACC 3<
Source B 9 9<

MOV
Move
Source 7 7<
Dest O:0.0 0<

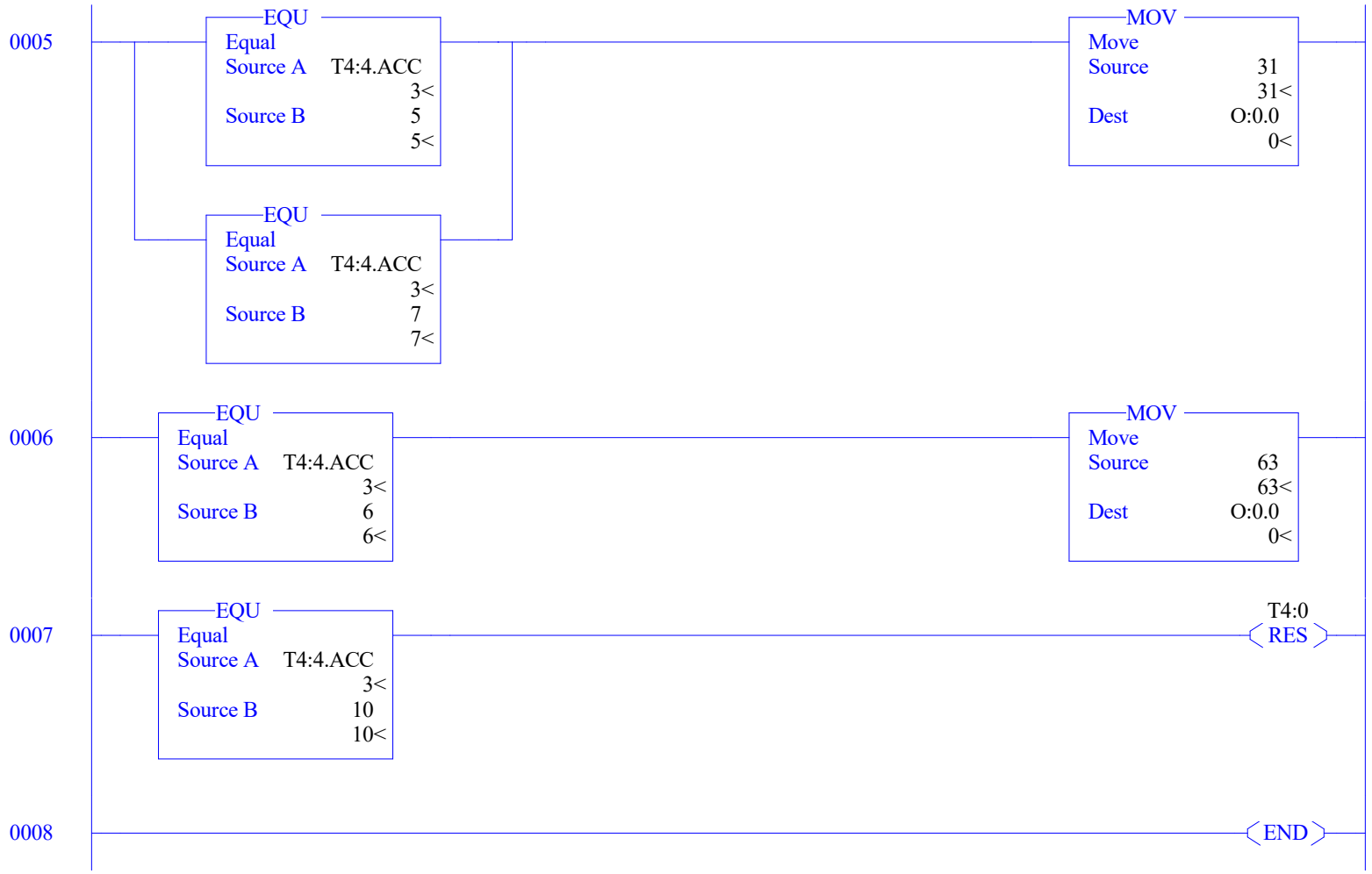
0004

EQU
Equal
Source A T4:4.ACC 3<
Source B 4 4<

EQU
Equal
Source A T4:4.ACC 3<
Source B 8 8<

MOV
Move
Source 15 15<
Dest O:0.0 0<

LAD 6 - SUB3 --- Total Rungs in File = 9



same as sub1 but leaving on the previous turned on lights

0000

TON
Timer On Delay
Timer T4:5
Time Base 1.0
Preset 7<
Accum 0<
EN
DN

turns on lights 3 and 4

0001

EQU
Equal
Source A T4:5.ACC
0<
Source B 1
1<

MOV
Move
Source 12
12<
Dest O:0.0
0<

turns lights 2 to 5

0002

EQU
Equal
Source A T4:5.ACC
0<
Source B 2
2<

MOV
Move
Source 30
30<
Dest O:0.0
0<

turns on all lights

0003

EQU
Equal
Source A T4:5.ACC
0<
Source B 3
3<

MOV
Move
Source 63
63<
Dest O:0.0
0<

same as rung 2

0004

EQU
Equal
Source A T4:5.ACC
0<
Source B 4
4<

MOV
Move
Source 30
30<
Dest O:0.0
0<

same as rung 1

0005

EQU
Equal
Source A T4:5.ACC
0<
Source B 5
5<

MOV
Move
Source 12
12<
Dest O:0.0
0<

turns off all lights; resets timer 5 to re-start the sequence again

0006

EQU
Equal
Source A T4:5.ACC
0<
Source B 6
6<

MOV
Move
Source 0
0<
Dest O:0.0
0<

T4:5
RES

0007

END

Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
O:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
I:0.0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 o

Data File S2 (hex) -- STATUS

Main

Processor Mode S:1/0 - S:1/4 = Remote Run
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 1100-1000-1000-0111

Proc

OS Catalog Number S:57 = 1200
OS Series S:58 = C
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = A
Processor FRN S:62 =
User Program Type S:63 = 4110h
Compiler Revision Number S:64 =

Scan Times

Maximum (x10 ms) S:22 = 16
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 6
Scan Toggle Bit S:33/9 = 1

Math

Math Overflow Selected S:2/14 = 0
Overflow Trap S:5/0 = 0
Carry S:0/0 = 0
Overflow S:0/1 = 0
Zero Bit S:0/2 = 1
Sign Bit S:0/3 = 0
Math Register (lo word) S:13 = 0
Math Register (high word) S:14-S:13 = 0
Math Register (32 Bit) S:14-S:13 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = Remote Run
Node Address S:15 (low byte) = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0
Outgoing Msg Cmd Pending S:33/2 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0
Startup Protection Fault S:1/9 = 0
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0
Control Register Error S:5/2 = 0
Major Error Executing User
Fault Rtn. S:5/3 = 0
Retentive Data Lost S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
Fault Routine S:29 = 0
Major Error S:6 = 0h
Error Description:

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- BINARY

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol)	Description
B3:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol)	Description
T4:0	0	0	0	1.0 sec	10	0		
T4:1	0	0	0	1.0 sec	5	0		
T4:2	0	0	0	1.0 sec	4	0		
T4:3	1	1	0	.01 sec	50	9		
T4:4	1	1	0	1.0 sec	10	3		
T4:5	0	0	0	1.0 sec	7	0		

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol)	Description
C5:0	0	0	0	0	0	0	8	1		reset to one (default)

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol)	Description
R6:0	0	0	0	0	0	0	0	0	0	0		

Data File N7 (dec) -- INTEGER

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	0									

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0				

Address (Symbol) = Value [Description]

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code	ABV	BI
C5:0			reset to one (default)				
C5:0.ACC							
I:0/0	SELECT	Global	Select subroutine				
I:0/1							
O:0.0							
O:0/0							
O:0/3							
S:0			Arithmetic Flags				
S:0/0			Processor Arithmetic Carry Flag				
S:0/1			Processor Arithmetic Underflow/ Overflow Flag				
S:0/2			Processor Arithmetic Zero Flag				
S:0/3			Processor Arithmetic Sign Flag				
S:1			Processor Mode Status/ Control				
S:1/0			Processor Mode Bit 0				
S:1/1			Processor Mode Bit 1				
S:1/2			Processor Mode Bit 2				
S:1/3			Processor Mode Bit 3				
S:1/4			Processor Mode Bit 4				
S:1/5			Forces Enabled				
S:1/6			Forces Present				
S:1/7			Comms Active				
S:1/8			Fault Override at Powerup				
S:1/9			Startup Protection Fault				
S:1/10			Load Memory Module on Memory Error				
S:1/11			Load Memory Module Always				
S:1/12			Load Memory Module and RUN				
S:1/13			Major Error Halted				
S:1/14			Access Denied				
S:1/15			First Pass				
S:2/0			STI Pending				
S:2/1			STI Enabled				
S:2/2			STI Executing				
S:2/3			Index Addressing File Range				
S:2/4			Saved with Debug Single Step				
S:2/5			DH-485 Incoming Command Pending				
S:2/6			DH-485 Message Reply Pending				
S:2/7			DH-485 Outgoing Message Command Pending				
S:2/15			Comms Servicing Selection				
S:3			Current Scan Time/ Watchdog Scan Time				
S:4			Time Base				
S:5/0			Overflow Trap				
S:5/2			Control Register Error				
S:5/3			Major Err Detected Executing UserFault Routine				
S:5/4			M0-M1 Referenced on Disabled Slot				
S:5/8			Memory Module Boot				
S:5/9			Memory Module Password Mismatch				
S:5/10			STI Overflow				
S:5/11			Battery Low				
S:6			Major Error Fault Code				
S:7			Suspend Code				
S:8			Suspend File				
S:9			Active Nodes				
S:10			Active Nodes				
S:11			I/O Slot Enables				
S:12			I/O Slot Enables				
S:13			Math Register				
S:14			Math Register				
S:15			Node Address/ Baud Rate				
S:16			Debug Single Step Rung				
S:17			Debug Single Step File				
S:18			Debug Single Step Breakpoint Rung				
S:19			Debug Single Step Breakpoint File				
S:20			Debug Fault/ Powerdown Rung				
S:21			Debug Fault/ Powerdown File				
S:22			Maximum Observed Scan Time				
S:23			Average Scan Time				
S:24			Index Register				
S:25			I/O Interrupt Pending				
S:26			I/O Interrupt Pending				
S:27			I/O Interrupt Enabled				
S:28			I/O Interrupt Enabled				
S:29			User Fault Routine File Number				
S:30			STI Setpoint				
S:31			STI File Number				
S:32			I/O Interrupt Executing				
S:33			Extended Proc Status Control Word				
S:33/0			Incoming Command Pending				
S:33/1			Message Reply Pending				
S:33/2			Outgoing Message Command Pending				
S:33/3			Selection Status User/DF1				
S:33/4			Communicat Active				
S:33/5			Communicat Servicing Selection				
S:33/6			Message Servicing Selection Channel 0				
S:33/7			Message Servicing Selection Channel 1				
S:33/8			Interrupt Latency Control Flag				
S:33/9			Scan Toggle Flag				
S:33/10			Discrete Input Interrupt Reconfigur Flag				
S:33/11			Online Edit Status				
S:33/12			Online Edit Status				
S:33/13			Scan Time Timebase Selection				

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code	ABV	BI
S:33/14			DTR Control Bit				
S:33/15			DTR Force Bit				
S:34			Pass-thru Disabled				
S:34/0			Pass-Thru Disabled Flag				
S:34/1			DH+ Active Node Table Enable Flag				
S:34/2			Floating Point Math Flag Disable,Fl				
S:35			Last 1 ms Scan Time				
S:36			Extended Minor Error Bits				
S:36/8			DII Lost				
S:36/9			STI Lost				
S:36/10			Memory Module Data File Overwrite Protection				
S:37			Clock Calendar Year				
S:38			Clock Calendar Month				
S:39			Clock Calendar Day				
S:40			Clock Calendar Hours				
S:41			Clock Calendar Minutes				
S:42			Clock Calendar Seconds				
S:43			STI Interrupt Time				
S:44			I/O Event Interrupt Time				
S:45			DII Interrupt Time				
S:46			Discrete Input Interrupt- File Number				
S:47			Discrete Input Interrupt- Slot Number				
S:48			Discrete Input Interrupt- Bit Mask				
S:49			Discrete Input Interrupt- Compare Value				
S:50			Processor Catalog Number				
S:51			Discrete Input Interrupt- Return Number				
S:52			Discrete Input Interrupt- Accumulat				
S:53			Reserved/ Clock Calendar Day of the Week				
S:55			Last DII Scan Time				
S:56			Maximum Observed DII Scan Time				
S:57			Operating System Catalog Number				
S:58			Operating System Series				
S:59			Operating System FRN				
S:61			Processor Series				
S:62			Processor Revision				
S:63			User Program Type				
S:64			User Program Functional Index				
S:65			User RAM Size				
S:66			Flash EEPROM Size				
S:67			Channel 0 Active Nodes				
S:68			Channel 0 Active Nodes				
S:69			Channel 0 Active Nodes				
S:70			Channel 0 Active Nodes				
S:71			Channel 0 Active Nodes				
S:72			Channel 0 Active Nodes				
S:73			Channel 0 Active Nodes				
S:74			Channel 0 Active Nodes				
S:75			Channel 0 Active Nodes				
S:76			Channel 0 Active Nodes				
S:77			Channel 0 Active Nodes				
S:78			Channel 0 Active Nodes				
S:79			Channel 0 Active Nodes				
S:80			Channel 0 Active Nodes				
S:81			Channel 0 Active Nodes				
S:82			Channel 0 Active Nodes				
S:83			DH+ Active Nodes				
S:84			DH+ Active Nodes				
S:85			DH+ Active Nodes				
S:86			DH+ Active Nodes				
T4:0							
T4:0.ACC	V	Global					
T4:0/DN							
T4:1							
T4:1.ACC							
T4:1/DN							
T4:2							
T4:2.ACC							
T4:3							
T4:3/DN							
T4:4							
T4:4.ACC							
T4:4/DN							
T4:5							
T4:5.ACC							
U:0							
U:1							
U:2							
U:3							
U:4							
U:5							
U:6							
U:7							

Instruction Comment Database

Address	Instruction	Description

Symbol Group Database

Group_Name Description